

SYSTEM & UML DIAGRAMS

AI Chatbot for College · BBA Computer Applications

1. Use Case Diagram

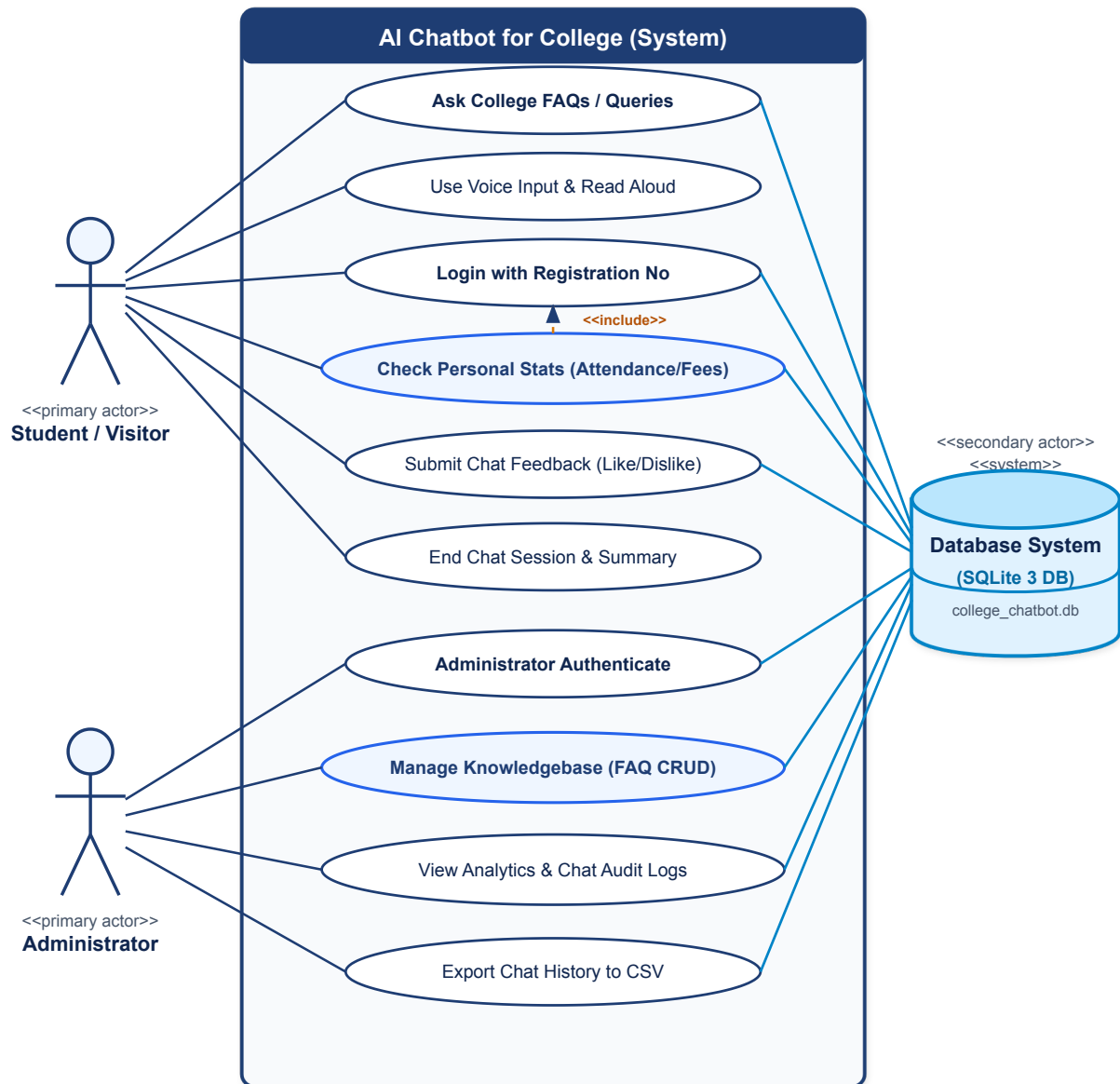


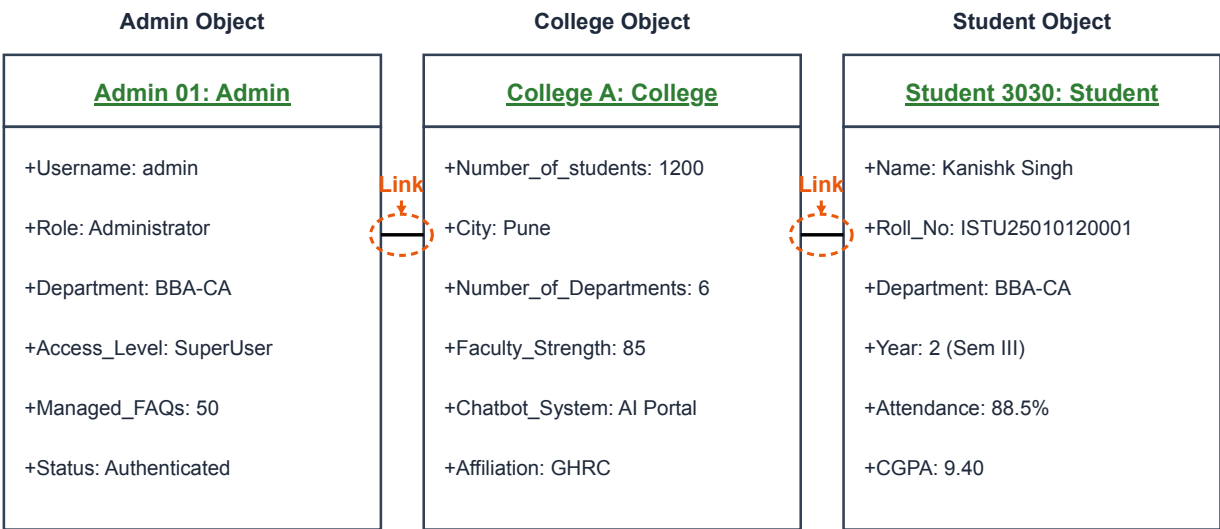
Fig 1: Use Case Diagram — primary actors (Student & Administrator), system boundary, and broad secondary database cylinder (SQLite 3 DB) with data-tier associations.

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3. Object Diagram (Runtime Instances & Slot Values)

An object diagram using links and 3 objects



An object of class Admin and an object of class Student are linked to an object of class College.

Detailed Object Diagram Breakdown (3 Concrete Instances):

- **Admin Object (Admin 01: Admin):** Models administrator authentication, department role, and active session permissions.
- **College Object (College A: College):** Central institutional node managing student capacity (1200), department quotas, and AI Chatbot.
- **Student Object (Student 3030: Student):** Concrete student instance Kanishk Singh (ISTU25010120001) with 88.5% attendance and CGPA.
- **Link Semantics:** Both Admin and Student are connected via formal binary links to the College entity, representing institutional association.

Fig 3: Object Diagram — runtime object instances (Admin Object, College Object, Student Object) connected via formal UML links matching GeeksforGeeks standard notation.

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6. Data Flow Diagram (DFD Level 0 Context & Level 1 Detailed)

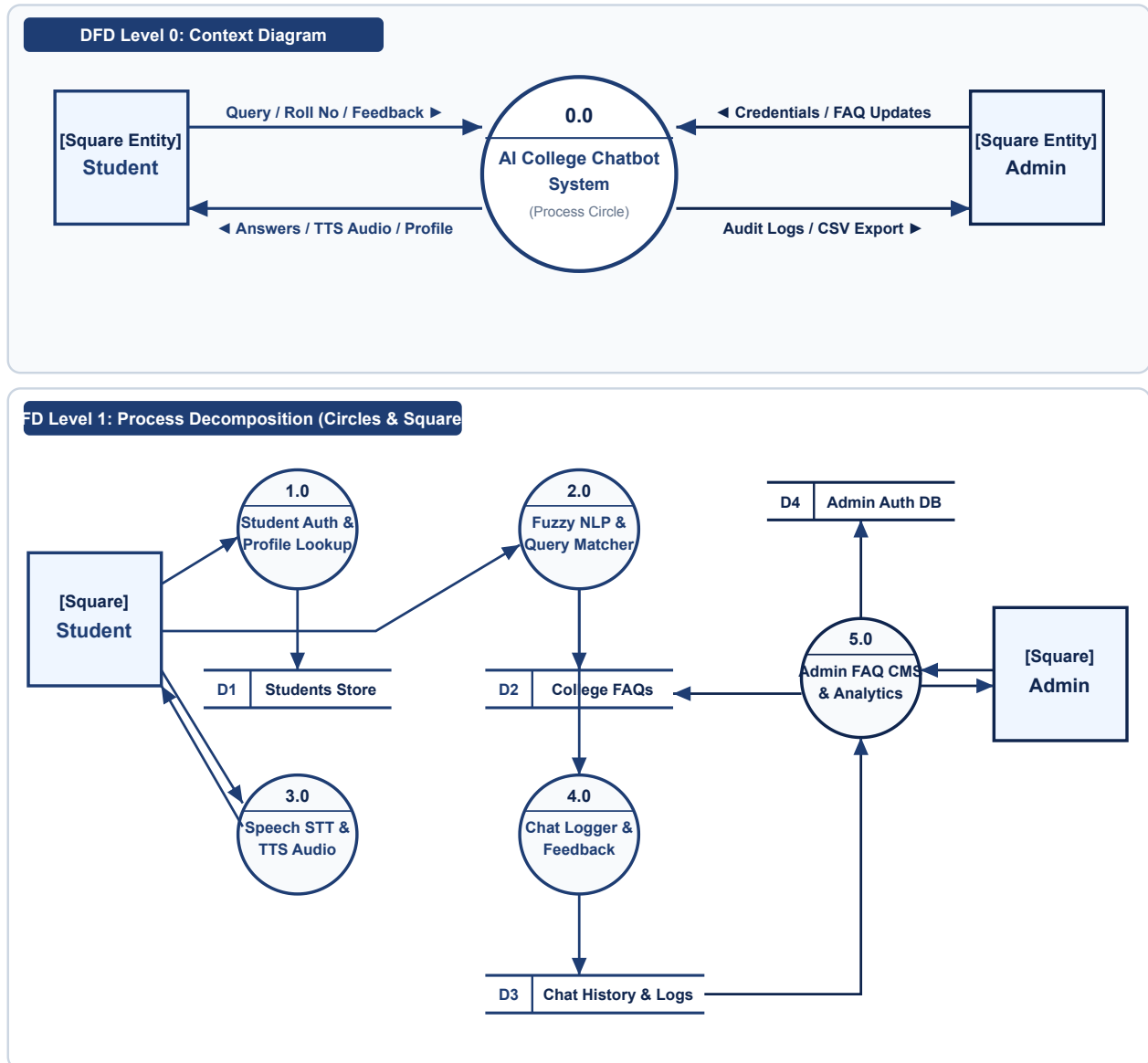
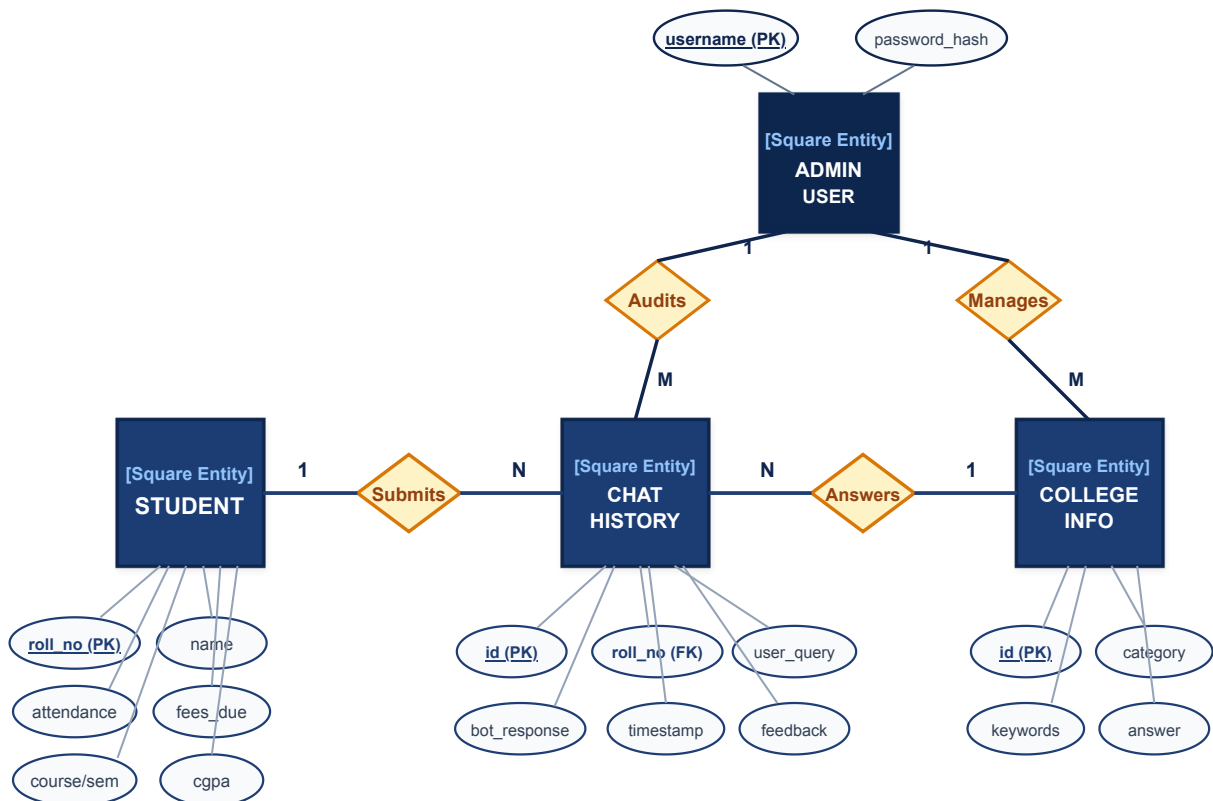


Fig 6: Data Flow Diagram (DFD) — Level 0 Context & Level 1 Process Breakdown using square entity symbols, circle process symbols, and solid data flow arrows.

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7. Entity-Relationship Diagram (ERD — Database Architecture)



✓ Standard Entity-Relationship Diagram Flow Summary:

- **Exact Square Entity Symbols:** All entities (STUDENT, ADMIN_USER, CHAT_HISTORY, COLLEGE_INFO) rendered as square boxes.
- **Left-to-Right Primary Data Flow:** STUDENT [1] submits [N] CHAT_HISTORY, which answers [N:1] from COLLEGE_INFO.
- **Top-to-Bottom Admin Control Flow:** ADMIN_USER [1] audits [M] CHAT_HISTORY and manages [M] COLLEGE_INFO FAQs.
- **Relational Integrity:** Primary Keys underlined (roll_no, id, username); Foreign Key marked (roll_no FK).

Fig 7: Entity-Relationship Diagram (ERD) — structured left-to-right relational flow using square entity symbols (STUDENT, ADMIN_USER, CHAT_HISTORY, COLLEGE_INFO) with cardinalities and non-crossing attributes.

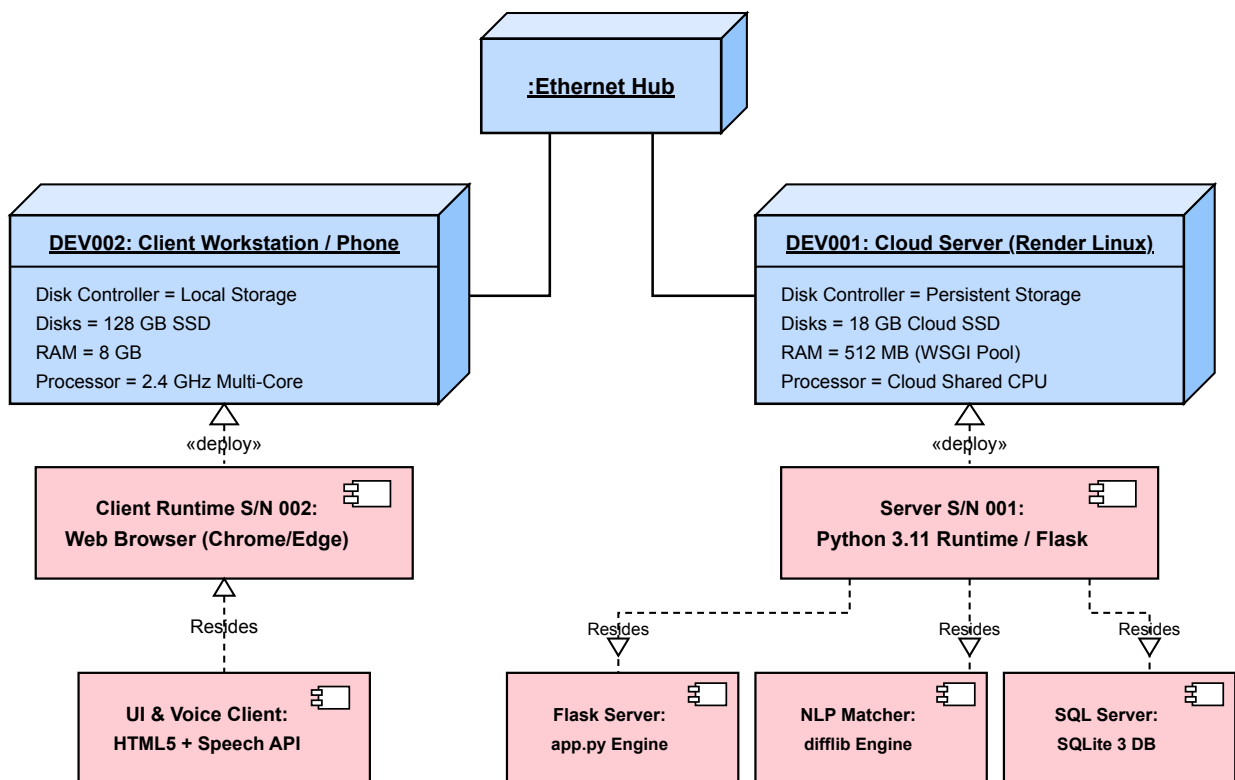
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8. Deployment Diagram (Hosting Topology & Artifact Architecture)

deployment Development Environment Model

Development Environment Model



UML Deployment Architecture Legend & Mapping:

- **:Ethernet Hub:** Central network routing gateway connecting client and cloud server hardware nodes.
- **3D Nodes (DEV001 & DEV002):** Physical/virtual hardware devices with CPU, RAM, and Disk controller specifications.
- **«deploy» & Resides:** Formal deployment relationships indicating software execution runtime and resident artifacts.

Fig 8: Deployment Diagram — Development Environment Model with :Ethernet Hub, 3D hardware nodes with specifications, «deploy» dependency arrows, and resident software component boxes.